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Orchestrator

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Global IJCSA: One Co-Simulation Step $t \rightarrow t+dt$

Collect ALL interface inputs U from zero-delay connections

Initialize $U^{(0)}$ from current input values

Phase 1: Global Interface-Newton ($dt=0$)

loop [until convergence ($m = 0, 1, 2, \dots$)]

1. Evaluate all components with frozen states ($dt=0$)

par [Parallel Execution]

evaluate_outputs($U1^{(m)}, t$)

$Y1^{(m+1)} = S1(U1^{(m)})$

$Y1^{(m+1)}$

evaluate_outputs($U2^{(m)}, t$)

$Y2^{(m+1)} = S2(U2^{(m)})$

$Y2^{(m+1)}$

evaluate_outputs($U3^{(m)}, t$)

$Y3^{(m+1)} = S3(U3^{(m)})$

$Y3^{(m+1)}$

2. Compute global residual

$R^{(m)} = [U1^{(m)} - Y_driver1^{(m+1)}] [U2^{(m)} - Y_driver2^{(m+1)}] [U3^{(m)} - Y_driver3^{(m+1)}]$

3. Check convergence

alt [|| $R^{(m)}$ || < tolerance]

✓ Converged

Exit iteration

4. Compute global Jacobian

$J^{(m)} = dR/dU$
via finite differences

Perturb each U_j ,
evaluate all outputs,
compute $J[:, j]$

5. Solve for correction

$J^{(m)} \cdot \Delta U^{(m)} = -R^{(m)}$

6. Apply correction

$U^{(m+1)} = U^{(m)} + \Delta U^{(m)}$

Phase 2: Time Integration with consistent interface

set_inputs($U1^*, t$)

committed

set_inputs($U2^*, t$)

committed

set_inputs($U3^*, t$)

committed

All components advance time

par [Parallel Time Integration]

do_step(t, dt)

Integrate $t \rightarrow t+dt$

$Y1(t+dt)$

do_step(t, dt)

Integrate $t \rightarrow t+dt$

$Y2(t+dt)$

do_step(t, dt)

Integrate $t \rightarrow t+dt$

$Y3(t+dt)$

All components at $t+dt$
Ready for next step

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